LARS-3D: Linear Global-Context Aggregation in Sparse Convolutional Networks for Robust Outdoor 3D Point Cloud Segmentation

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***Abstract*— Recent advances in LiDAR semantic segmentation have largely focused on structured urban driving datasets, where sparse convolutional and attention-based models achieve strong performance. However, their behavior in unstructured, terrain-rich environments remains underexplored. In this work, we investigate semantic segmentation in complex outdoor park and off-road scenes characterized by irregular terrain, dense vegetation, and fine-grained ground categories. To address this challenge, we propose LARS-3D, a novel Linear Global-Context Aggregation model for Robust Semantic Segmentation in 3D, integrating a linear global-context aggregation module within a sparse convolutional backbone. We also introduce EastPark, a novel large-scale multi-sensor dataset comprising over 9,000 annotated LiDAR scans with 22 terrain-aware semantic classes. We acquired and manually annotated it with a UGV to benchmark perception methods in outdoor, unstructured, complex public environments. Our analysis reveals substantial cross-domain degradation between urban-trained models and terrain- and vegetation-rich scenes. Extensive experiments demonstrate consistent improvements over strong baselines on EastPark, RELLIS-3D, and SemanticPOSS, achieving competitive performance across unstructured outdoor and structured urban benchmarks. We provide a detailed ablation study showing that the global-context aggregation module improves the performance for outdoor 3D semantic segmentation. The implementation of LARS-3D and our EastPark dataset are at `https://lars-3d.github.io`**

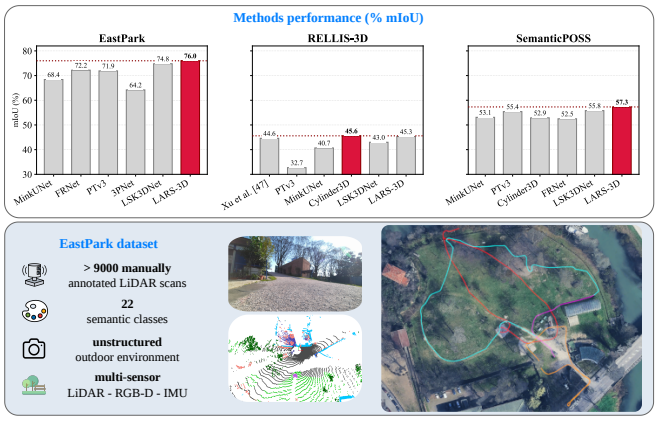


Fig. 1: Overview of our contributions. Top: LARS-3D, a novel method for Robust Segmentation in 3D outdoor environment, achieving competitive performance with linear global context aggregation. Below: EastPark, a multi-sensor, large-scale dataset acquired in a natural park. We overlaid trajectories on a Google Satellite Image (©2025 Google). Sequence 00 ■, Sequence 01 ■, Sequence 02 ■, Sequence 03 ■ (best viewed in color).

I. INTRODUCTION

Semantic segmentation of LiDAR point clouds is a fundamental component of autonomous navigation and robotic perception that aims to assign a semantic label to each point in a 3D scan. Recent progress has been driven largely by urban driving benchmarks such as SemanticKITTI [1] and nuScenes [2], where sparse convolutional and attention-based architectures achieve impressive performance. However, these datasets predominantly capture structured road environments with planar surfaces, well-defined object categories, and limited terrain variability. In contrast, real-world outdoor environments such as natural parks, forests, and off-road areas exhibit highly irregular terrain, dense vegetation, numerous fine-grained ground categories, and frequent geometric discontinuities. Models trained on structured urban data ([3], [4], [5], [6]) often struggle in these settings [7], [8], revealing a substantial cross-domain degradation between urban benchmarks and vegetation- and terrain-rich environments.

To systematically study this problem, we acquired and manually annotated EastPark, a large-scale multi-sensor dataset collected in an unstructured, outdoor, and complex public park environment. It comprises over 9,000 manually annotated LiDAR scans spanning 22 semantic classes, including detailed terrain categories such as grass, soft gravel, hard gravel, mud, and rubber surfaces. The dataset also includes RGB-D images and IMU data. Our analysis shows that state-of-the-art models pre-trained on urban data experience significant performance degradation when evaluated on EastPark, particularly for vegetation and terrain classes. We argue that a key limitation of existing approaches lies in their local receptive fields or quadratic-complexity global attention mechanisms. While sparse convolutions effectively capture local geometry, modeling long-range contextual dependencies remains challenging. Self-attention addresses global reasoning but incurs quadratic complexity and significant memory overhead, limiting scalability for dense point clouds.

In this work, we propose LARS-3D, a novel linear global context aggregation (GCA) model for robust 3D point cloud semantic segmentation. By injecting a linear-complexity

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global context inspired by RWKV [9] (Receptance Weighted Key Value) within a sparse convolutional backbone, our method captures long-range dependencies while maintaining linear complexity with respect to the number of points. This design allows for both local geometric expressiveness and global reasoning without the computational burden of full quadratic self-attention. We evaluate our approach on EastPark and RELLIS-3D, two benchmarks characterized by unstructured outdoor environments, but also on SemanticPOSS, characterized by urban scenes, to demonstrate the generalization of our method across different domains, achieving competitive performance in all the datasets. We further provide comprehensive ablations analyzing model depth and computational efficiency, demonstrating the robustness and scalability of GCA modeling for large-scale 3D perception. Our contributions, highlighted in Fig.1, are threefold: (i) We propose LARS-3D, a novel linear global context aggregation model for robust 3D semantic segmentation, enabling efficient long-range contextual reasoning; (ii) We further introduce EastPark, a large-scale, multi-modal, terrain-rich dataset for evaluating semantic segmentation in unstructured outdoor environments, comprising over 9,000 manually annotated LiDAR scans across 22 classes, RGB-D images and IMU data; (iii) We demonstrate strong and competitive performance across multiple unstructured and structured outdoor benchmarks, and provide analyses of efficiency and architectural design.

II. RELATED WORK

A. 3D LiDAR Semantic Segmentation

Recent methods in 3D LiDAR semantic segmentation can be categorized into three main approaches: point-based, projection-based, and voxel-based.

Point-based methods are a direct approach to processing raw point clouds, a direction established by pioneering methods that introduced techniques to handle unstructured data by learning hierarchical local features [10], [11], [12], [13]. More recently, there has been a marked shift towards transformer-based architectures, which have proven to be highly effective in modeling long-range dependencies. The Point Transformer series [14], [15], [4] exemplifies this trend, with Point Transformer v2 [15] enhancing the self-attention mechanism and the latest iteration, Point Transformer v3 [4], achieving state-of-the-art performance through a simplified and more efficient design that scales to larger receptive fields. Similarly, Swin3D [16] adapted the Swin Transformer, a very effective method in 2D, for 3D, thereby demonstrating the efficacy of pretrained backbones in facilitating scene understanding.

Projection-based methods leverage the efficiency of 2D convolutions by first transforming the sparse 3D point cloud into dense 2D representations. A prevalent technique involves the projection of points onto a spherical or range view, as exemplified by [17], [18], [19]. RangeNet++ [20] adopts DarkNet to process range images and proposes an efficient GPU-based k-NN post-processing technique. More recent work, such as [3], has pushed the efficiency of this paradigm by demonstrating that competitive performance can be achieved with simple MLP and 2D convolution architectures. Furthermore, a recent trend involves creating hybrid models, such as 3PNet [21], that combine the speed of range images with the detail of local point-based features, proving particularly effective in low-data scenarios. This highlights the central trade-off of projection methods: gaining significant computational speed at the potential cost of fine-grained geometric fidelity.

Voxel-based methods operate by quantizing point clouds into a regular 3D grid, making them compatible with 3D convolutional neural networks. Initially, this approach was constrained by the prohibitive computational and memory demands of dense 3D convolutions on largely empty voxel spaces (e.g., in the pioneering work [22]). A pivotal breakthrough came with the introduction and optimization of sparse convolutions, facilitating convolution operations only on non-void voxels [23], [24]. Subsequent research has focused on both adapting the voxel representation and enhancing the network architectures. Methods such as Cylinder3D [6] or $(AF)^2$-S3Net [25] introduced cylindrical and asymmetrical convolutions that align with the sensor's natural geometry and incorporated attentive feature fusion and adaptive selection mechanisms respectively. More recent work focuses on increasing the receptive field to capture more context. Thus, LargeKernel3D [26] concluded that replacing stacks of small 3D sparse convolutions with a single larger kernel significantly improves performance by expanding the receptive field. Building directly upon this concept, LSK3DNet [5] introduces a more advanced framework to make these large kernels both effective and efficient. It employs dynamic sparsity and channel pruning to learn an optimized sparse kernel structure during training.

B. Fast and Efficient Transformer Alternatives

Transformers [27] have driven significant progress across numerous domains through the introduction of the attention mechanism, delivering powerful and reliable performance. However, their practical deployment is often constrained by the quadratic computational and memory costs of self-attention. To address this limitation, several linear-complexity alternatives have been proposed. Linear Attention [28] replaces the softmax operation with linear kernel approximations to reduce complexity. AFT [29] introduces element-wise operations for more efficient computation. Retentive Networks [30] explicitly capture temporal dependencies using a decay matrix, achieving linear complexity. More recently, RWKV [9] reformulates attention within an RNN-based framework, while Mamba [31] uses a state-space formulation, both achieving linear complexity. Such sequence-based models have also been applied to point clouds [32], [33], but their efficient application to large-scale outdoor point clouds remains underexplored. Unlike these recurrent/state-space approaches, our method uses linear GCA within a sparse point-voxel architecture for large-scale outdoor semantic segmentation.

TABLE I: Comparison of recent perception datasets.

Dataset	Year	Scene	Platform	Sensors					Scale
				LiDAR	Depth	RGB	IMU	3D ann.	
Structured environments (on-road / campus)									
SemanticKITTI	2019	Urban	Vehicle	✓	–	✓	✓	✓	Large
SemanticPOSS	2020	Urban	Vehicle	✓	–	✓	✓	✓	Large
nuScenes	2020	Urban	Vehicle	✓	–	✓	✓	✓	Large
Waymo Open	2020	Urban	Vehicle	✓	–	✓	✓	✓	Large
PandaSet	2021	Urban	Vehicle	✓	–	✓	✓	✓	Large
Unstructured environments (off-road / natural)									
YCOR	2018	Forest	UGV	–	–	✓	–	–	Small
RUGD	2019	Off-road	UGV	✓	–	✓	✓	–	Small
RELLIS-3D	2020	Off-road	UGV	✓	✓	✓	✓	✓	Large
GOOSE	2024	Off-road	Vehicle	✓	–	✓	✓	✓	Large
BotanicGarden	2024	Nat. park	UGV	✓	–	✓	✓	–	Small
WildScenes	2025	Forest	UGV	✓	–	✓	✓	✓	Large
EastPark (ours)	2026	Nat. park	UGV	✓	✓	✓	✓	✓	Large

C. Public UGV Datasets

3D LiDAR semantic segmentation is a fundamental aspect of environmental perception for autonomous robots, enabling safe navigation by providing a detailed point-wise understanding of the surroundings. The public release of extensive on-road traffic datasets has actively contributed to the constant improvement of current perception models. Pioneering benchmarks such as SemanticKITTI [1] and SemanticPOSS [34], together with nuScenes [2], PandaSet [35], and the Waymo Open Dataset [36], are characterized by their well-defined road, building, and vehicle traffic scenarios. Other datasets have been collected by unmanned ground vehicles (UGVs) in structured environments. An example is the NCLT dataset [37] collected by a Segway UGV, focused on a university campus for long-term autonomy. Another example is the M2DGR dataset [38], providing multi-modal data from an AgileX Scout Mini for robot navigation through structured indoor and outdoor campus areas.

To enable robust robotic autonomy, datasets capturing unstructured and natural environments are indispensable. These are acquired using various platforms, including specialized vehicles [39], handheld sensor rigs [40], and, most prevalently, UGVs. The focus on UGVs is critical, as their primary application domain is in these unstructured settings, where perception models trained exclusively on structured data experience a substantial decline in performance. This failure stems from a significant domain gap: structured scenes are characterized by planar surfaces and rigid objects, while unstructured terrains are marked by highly irregular ground, diverse vegetation (e.g., tall grass, bushes), and natural obstacles absent from urban or campus data.

In recent years, a new generation of UGV-centric datasets has emerged to replicate the success of perception models in these unstructured outdoor environments. RELLIS-3D [41] has been a pivotal contribution to full-scene semantic understanding by providing densely annotated data from an off-road campus setting, complete with two LiDARs (32-beam and 64-beam), cameras, and IMU data. On a smaller scale, the Botanic Garden dataset [42] also addresses full-scene description in a park-like environment; however, it only provides vision-based (camera/RGB) semantic annotation, without manually annotated LiDAR semantic labels.

In contrast to these full-scene benchmarks, other datasets focus on more specific perception challenges. For instance, the RUGD dataset [43] focuses exclusively on the classification of rugged ground surfaces such as dirt, grass, and rocks for traversability analysis. Similarly, the YCOR dataset [44] targets the specific task of trail navigation in rugged terrain, providing labels for narrow, challenging paths characterized by rocky and heavily vegetated surroundings.

While these datasets have been instrumental in advancing off-road perception, a gap remains for a large-scale, multi-sensor dataset captured in a diverse, public natural park that integrates well-maintained trails, wilderness, and various types of terrains. Tab. I compares our work with key state-of-the-art datasets on 3D semantic segmentation. Our dataset combines a rich sensor suite with high-quality, manual 3D annotations, providing high data fidelity and fine-grained labeling. These features make EastPark a valuable resource for advancing perception research in mobile robotics.

III. EastPark Dataset

Our EastPark dataset consists of 9056 annotated 3D point clouds, 13584 RGB-D images, and circa 30K IMU readings. Point clouds are provided in a raw per-point format, including Cartesian coordinates (*x, y, z*), *remission*, and *timestamp*. Each point is further augmented with *reflectivity*, i.e., a calibrated value indicating the material properties and texture of the scanned surface, *ring*, the index of the vertical laser channel that generated the point and *ambient*, a measurement of ambient near-infrared light, which can be used to identify and filter noise from external light sources like sunlight.

We collected four sequences, with 3120, 1184, 2607, and 2145 point clouds, respectively, with a UGV. We acquired all point clouds under uniform weather conditions on a clear, sunny day in mid-April 2025. Although the initial three sequences are quite similar in terms of label distribution, the fourth sequence exhibits a higher degree of heterogeneity. This is due to the fact that the UGV was partially driven over a sidewalk along the road that limits the park, resulting in the encounter of classes that are not observed in the other sequences. Consequently, the scans of the fourth sequence were randomly assigned, one by one, to three partitions (sequences 3, 4 and 5), so that its classes appear in training, validation and test. As a result, sequence 5 contains frames temporally adjacent to training frames and is not an independent test set; road, sidewalk, rubber, car, bicycle and motorcycle occur only in this drive and are evaluated on it alone. We therefore also report sequence 2, a separate drive, as held-out test (Tab. II). Finally, the dataset consists of six sequences totaling 3120, 1184, 2607, 1073, 429, and 643 point clouds. Sequences 0 and 3 (4193 scans, 46%) are used for training, sequences 1 and 4 (1613 scans, 18%) for validation, and sequences 2 and 5 (3250 scans, 36%) for testing. Since LiDAR and camera have different frame rates, we provide RGB-D images with timestamps synchronized to the LiDAR and IMU data, allowing straightforward temporal alignment across all sensors for

multi-modal perception tasks. We performed sensor calibration offline by manually identifying corresponding reference points between camera images and LiDAR point clouds across multiple frames, followed by a joint optimization to estimate the extrinsic transformation matrix. We manually annotated the dataset with 22 classes. Two human annotators first performed a global annotation by aggregating all point clouds using ego-motion estimates obtained from visual odometry. This initial annotation was subsequently refined on a per-scan basis to account for temporal misalignments and scene-specific ambiguities. Both annotators independently inspected each scan multiple times, after which ambiguous or inconsistent cases were jointly reviewed and corrected. To address privacy concerns, faces in image streams were blurred. The class distribution exhibits a natural imbalance, with higher frequencies for common static structures and lower representation for dynamic or smaller classes such as motorcycles, kids, and bicycles. For semantic segmentation tasks, we annotated a special class, called *follower*, i.e. the people maneuvering the UGV. We merged this class with the class *person* because it denotes the operational role of a human rather than a distinct semantic object category. Beyond approximately 30 m, manual labeling becomes increasingly unreliable because objects are represented by only a small number of noisy LiDAR returns. To ensure consistent ground truth quality, we limit the annotated region to 30 m.

IV. LARS-3D

Our goal is to enhance 3D semantic segmentation in unstructured outdoor environments by incorporating global contextual reasoning into sparse convolutional backbones. We build upon LSK3DNet [5] and redesign its global reasoning branch through a linear-complexity global context aggregation formulation, yielding LARS-3D, a unified framework that captures long-range dependencies while preserving the efficiency of sparse convolutional processing.

A. Problem Formulation

Let a LiDAR point cloud of N points be defined as $\mathbf{P} = \{(\mathbf{p}_i, \mathbf{f}_i)\}_{i=1}^{N}$, where $\mathbf{p}_i \in \mathbb{R}^3$ is the 3D coordinate of the i-th point, and $\mathbf{f}_i \in \mathbb{R}^C$ is the input feature vector (comprising, e.g., $\mathbf{p}_i$ and remission). We define $\mathbf{P}_c = \{\mathbf{p}_i\}_{i=1}^{N}$ and $\mathbf{P}_f = \{\mathbf{f}_i\}_{i=1}^{N}$. The point cloud semantic segmentation task aims to predict a class label $y_i \in \{1, \ldots, K\}$ for each point, with K the number of classes. We propose a novel method, LARS-3D, based on 3D sparse convolutions and linear global context aggregation (GCA), jointly processing features at voxel and point levels.

B. LARS-3D

LARS-3D architecture is depicted in Fig. 2. It is composed of a sequence of Sparse Point-Voxel (SPV) blocks [45] that process sparse voxel features, and point features hierarchically at multiple scales. Sparse voxels are processed by Large Sparse Kernel (LSK) blocks [5] with 3D sparse convolution, while point features are processed by the point-branch encoder implementing our linear GCA module.

The input point cloud $\mathbf{P}$ is preprocessed by the voxelizer module, which quantizes the point coordinates using voxel size v_s and extracts features at point level using linear projections, obtaining $\mathbf{P}^{(1)}$. At hierarchical level $s \in \{1, \ldots, S\}$, the s-th SPV block receives point cloud $\mathbf{P}^{(s)}$. We compute a sparse voxel lattice using the quantized point cloud by averaging point features $\mathbf{P}_f^{(s)}$ inside each voxel, obtaining $\mathbf{V}^{(s)} = \{\mathbf{V}_c^{(s)}, \mathbf{V}_f^{(s)}\}$, i.e. voxel lattice and features.

The voxel branch of each SPV block consists of two sparse 3D convolution residual blocks, which process $(\mathbf{V}_c^{(s)}, \mathbf{V}_f^{(s)})$ and produce updated voxel features $\mathbf{V}_{f,\text{vox}}^{(s)} = \{\mathbf{h}_{j,\text{vox}}^{(s)}\}_{j=1}^{N_v}$, where N_v is the number of voxels. Let $\psi(i)$ denote the mapping from point i to its corresponding voxel. The voxel features are then propagated to the points by feature fusion:

$$\mathbf{f}_{i,\text{vox}}^{(s)} = \mathbf{f}_i^{(s)} + \mathbf{h}_{\psi(i),\text{vox}}^{(s)}, \qquad i = 1, \ldots, N, \tag{1}$$

yielding the enriched point features $\mathbf{P}_{f,\text{vox}}^{(s)}$.

The point branch consists of our hierarchical linear GCA module, described in Sec IV-C:

$$\mathbf{P}_{f,\text{GCA}}^{(s)} = \text{GCA}_{\text{encoder}}^{(s)}(\mathbf{P}_c^{(s)}, \mathbf{P}_{f,\text{vox}}^{(s)}). \tag{2}$$

The outputs of the two branches are fused together at the point level. Once again, voxel features are added to the $\text{GCA}_{\text{encoder}}$ point features, obtaining the output of the s-th SPV block $\mathbf{P}^{(s+1)} = \{(\mathbf{p}_i^{(s)}, \mathbf{f}_i^{(s+1)})\}_{i=1}^{N}$, with

$$\mathbf{f}_i^{(s+1)} = \mathbf{f}_{i,\text{GCA}}^{(s)} + \mathbf{h}_{\psi(i),\text{vox}}^{(s)}. \tag{3}$$

The fused representation is passed to the next SPV block.

Finally, we concatenate all the point features at each scale s and feed the concatenated feature tensor to a classification head for point-wise semantic prediction.

C. Global Context Aggregation (GCA) Encoder

The GCA encoder introduces global context modeling within each SPV block with linear complexity. It captures long-range dependencies across the entire point cloud, which is crucial for understanding complex scenes in unstructured environments.

Our GCA encoder, depicted in Fig 2, lower-left, consists of: (i) a linear projection branch, for residual connection; (ii) a downsampling layer to reduce the number of points and increase the receptive field followed by projection; (iii) L stacked GCA blocks; (iv) an upsampling layer to restore the original point resolution; (v) a final feature fusion and linear mapping layer.

The downsampling layer merges points by scaled voxel coordinates to reduce sequence length and stabilize learning. Let $\delta(i)$ denote the mapping from point i to its corresponding scaled voxel. The new point coordinate of the i-th point $\mathbf{p}^{\star(s)}_i$ is given by:

$$\mathbf{p}^{\star(s)}_i = \delta(\mathbf{p}_i) = \lfloor \mathbf{p}_i / s \rfloor. \tag{4}$$

We merge points with the same scaled voxel coordinate by averaging their features $\mathbf{f}_{i,\text{vox}}^{(s)}$, obtaining the downsampled point features $\mathbf{X}^{(s)}$, containing $N_s << N$ elements. Then, we

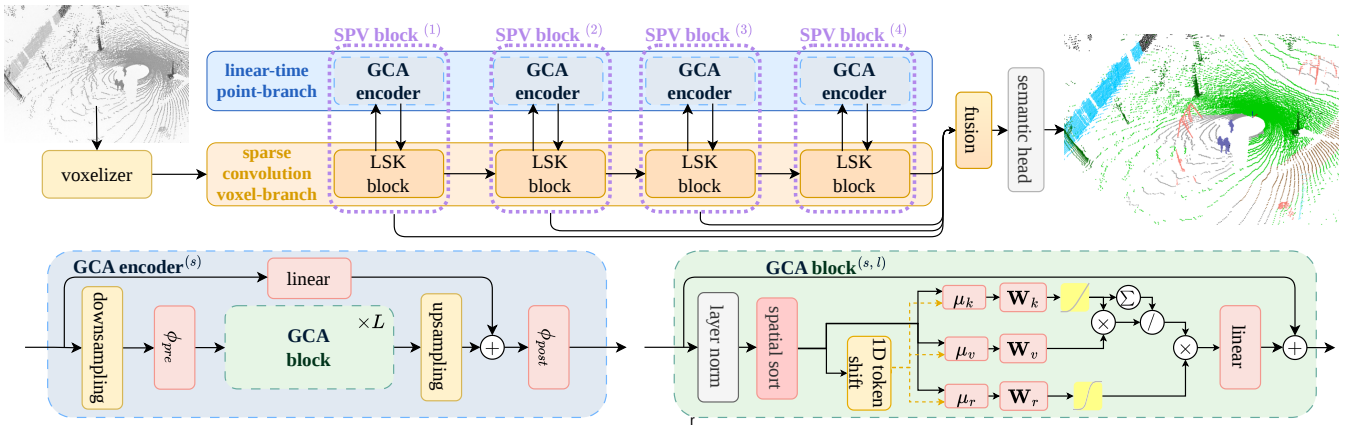


Fig. 2: Overview of our proposed approach (top) and of our GCA encoder (bottom left) and GCA block (bottom right). The input point cloud is preprocessed by the voxelizer module, which first quantizes the point coordinates and then extracts features at point level using linear projections. The SPV block processes these features at progressively coarser scales. At each scale, the voxel branch uses sparse 3D convolutions and the point branch employs our GCA encoder to capture global context. The outputs are fused and refined through multiple SPV blocks for hierarchical feature learning. A final classification head infers semantic prediction.

apply $\phi_{pre}^{(s)}$, a linear projection with bias to match the feature dimension expected by the GCA block:

$$\mathbf{X}_{\text{expanded}}^{(s)} = \phi_{pre}^{(s)}(\mathbf{X}^{(s)}). \tag{5}$$

The core of our GCA point encoder is the stack of L GCA blocks, which model global context with linear complexity. A GCA block begins with layer normalization of the input features:

$$\mathbf{X}_{\text{norm}}^{(s)} = \text{LayerNorm}(\mathbf{X}_{\text{expanded}}^{(s)}). \tag{6}$$

Since the token-shift operation requires an ordered token stream, we impose a deterministic spatial traversal. We use 3D Z-order (Morton order) to preserve spatial locality, so neighboring voxels are more likely to remain close in the resulting sequence: $\mathbf{X}_{\text{ordered}}^{(s)} = \pi(\mathbf{X}_{norm}^{(s)})$.

To incorporate local information from the preceding token in the spatial traversal and enable the GCA blocks to capture dependencies across the entire point cloud, we compute a shifted version of the input features, with $j = 1, \ldots, N_s$:

$$\mathbf{X}_{j,\text{shift}}^{(s)} = \begin{cases} \mathbf{0}, & j = 1 \\ \mathbf{X}_{j-1,\text{ordered}}^{(s)}, & j > 1. \end{cases} \tag{7}$$

Then, we compute three mixed versions of $\mathbf{X}_{\text{ordered}}^{(s)}$ and $\mathbf{X}_{\text{shift}}^{(s)}$, using learnable scalar parameters μ_k, μ_v and μ_r:

$$\begin{aligned} \mathbf{X}_k &= \mu_k \mathbf{X}_{\text{ordered}}^{(s)} + (1-\mu_k)\mathbf{X}_{\text{shift}}^{(s)}, \\ \mathbf{X}_v &= \mu_v \mathbf{X}_{\text{ordered}}^{(s)} + (1-\mu_v)\mathbf{X}_{\text{shift}}^{(s)}, \\ \mathbf{X}_r &= \mu_r \mathbf{X}_{\text{ordered}}^{(s)} + (1-\mu_r)\mathbf{X}_{\text{shift}}^{(s)}. \end{aligned} \tag{8}$$

To compute key, value and receptance we apply independent linear projections. Only receptance uses bias $\mathbf{b}_r$, while $\mathbf{W}_k$, $\mathbf{W}_v$, and $\mathbf{W}_r$ are the weight matrices, respectively, softplus is a smooth, differentiable approximation of the ReLU and σ is the sigmoid function (we omit $^{(s)}$ for readability):

$$\begin{aligned} \mathbf{K} &= \text{softplus}(\mathbf{W}_k \mathbf{X}_k), \\ \mathbf{V} &= \mathbf{W}_v \mathbf{X}_v, \\ \mathbf{R} &= \sigma(\mathbf{W}_r \mathbf{X}_r + \mathbf{b}_r). \end{aligned} \tag{9}$$

Instead of quadratic attention, we efficiently aggregate global contextual statistics across all tokens:

$$\begin{aligned} \mathbf{C}_{\mathbf{KV}} &= \sum_j \mathbf{K}_j \odot \mathbf{V}_j, \\ \mathbf{C}_{\mathbf{K}} &= \sum_j \mathbf{K}_j. \end{aligned} \tag{10}$$

We then compute a normalized global context vector and modulate each token using its point-wise gate:

$$\mathbf{C}_i = \frac{\mathbf{C}_{\mathbf{KV}}}{\mathbf{C}_{\mathbf{K}} + \varepsilon} \odot \mathbf{R}_i, \tag{11}$$

where $\varepsilon \in \mathbb{R}$ is a small constant to prevent division by zero.

We apply a final linear projection with bias to produce the output features of the GCA block, with skip-connection:

$$\mathbf{X}_{\text{out, ordered}}^{(s)} = (\mathbf{W}_o \mathbf{C} + \mathbf{b}_o)^{(s)} + \mathbf{X}_{\text{ordered}}^{(s)} \tag{12}$$

Inverse permutation restores original spatial order:

$$\mathbf{X}_{\text{out}}^{(s)} = \pi^{-1}(\mathbf{X}_{\text{out, ordered}}^{(s)}). \tag{13}$$

The GCA encoder stacks L consecutive GCA blocks, iteratively processing the downsampled point features, eventually producing $\mathbf{X}_{\text{out}}^{(s,L)}$. We map features back to original full resolution using the inverse map:

$$\mathbf{X}_{\text{full}}^{(s)} = \delta^{-1}(\mathbf{X}_{\text{out}}^{(s,L)}). \tag{14}$$

We apply skip-connection by concatenating with the identity branch and applying a linear projection:

$$\mathbf{P}_{f,\text{GCA}}^{(s)} = \phi_{\text{post}}^{(s)}([\mathbf{X}_{\text{full}}^{(s)}; \mathbf{X}^{(s)}]), \tag{15}$$

where $\phi_{post}^{(s)}$ is composed of a linear layer followed by LeakyReLU activation and another linear layer and $[\cdot,\cdot]$ is the concatenation operator. The stacking of GCA blocks is the key to obtain progressive spatial information propagation and long-range feature interaction. The computational complexity of a GCA block can be modeled as $O(N_s C^2)$, with a total complexity for the GCA encoder of $O(L N_s C^2)$, which is linear in the number of input tokens N_s.

V. EXPERIMENTS AND RESULTS

We conducted experiments to validate our claims that (i) the integration of global context through our GCA point encoder considerably enhances semantic segmentation performance in unstructured outdoor environments, and (ii) our approach achieves this improvement with linear computational complexity, making it efficient for real-world applications.

A. Datasets

We evaluate our approach on three challenging LiDAR semantic segmentation benchmarks spanning unstructured outdoor and urban environments. **RELLIS-3D** [41] is an off-road dataset designed for autonomous navigation in natural terrain. It contains LiDAR scans captured in forests, grasslands, and uneven ground surfaces, with annotations for terrain, vegetation, and structural objects. Compared to urban benchmarks, RELLIS-3D exhibits significantly higher geometric irregularity and class imbalance. We follow the official train/validation/test splits. **EastPark (ours)** is described in detail in Section III. **SemanticPOSS** is designed for semantic segmentation in autonomous driving, particularly focusing on urban, complex, crowded, and dynamic environments. It includes a wide variety of classes such as pedestrians, vehicles, and urban infrastructure, with a high density of points and significant occlusions. SemanticPOSS provides no separate test set; we therefore report its validation split in both the ablation and the final test, and fixed $L = 8$ in advance rather than selecting it on this split.

B. Implementation Details and Metrics

Training Parameters We implement our model in PyTorch, and train it from scratch using three NVIDIA L40S GPUs for 100 epochs, with batch size 8 for each dataset. We adopt AdamW optimizer, and cosine annealing scheduler with linear warm up. The learning rate increases to 1×10^{-3} in the first 5 epochs and then decreases to 1×10^{-5}, with weight decay of 0.05. The loss is a combination of cross-entropy and Lovász [46], with equal weights. We use standard data augmentation techniques, including rotation, flip and scale.
Network Parameters The network has $C = 4$ input features (x, y, z, and remission). We maintain the same network configuration of LSK3DNet, with $S = 4$ SPV stages with increasing scale (2, 4, 8, 16) but fixed channel dimension (64). We also adopted the same training strategy as LSK3DNet, which includes sparse kernel pruning and growing, and channel-wise weight selection. We stack $L = 8$ GCA layers.
Metrics We follow standard evaluation in semantic segmentation and utilize intersection-over-union (IoU) for each class c, reporting the average score (mIoU) over all classes, with $\mathrm{IoU}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i+\mathrm{FP}_i+\mathrm{FN}_i}$, where $\mathrm{TP}_i, \mathrm{FP}_i, \mathrm{FN}_i$ are the numbers of true positive, false positive and false negative respectively.

C. Cross-domain Degradation Analysis on EastPark

We first evaluate several state-of-the-art LiDAR semantic segmentation models on the proposed EastPark dataset using pre-trained weights without retraining. Table II reports results for models trained on existing large-scale datasets and evaluated directly on EastPark (vanilla), as well as results obtained through linear probing.

The results reveal a severe cross-domain degradation between existing LiDAR benchmarks and terrain-rich environments such as EastPark. When evaluated without adaptation, all vanilla models exhibit extremely low performance, with mIoU below 8%. This indicates that models trained on urban driving datasets fail to generalize to complex park environments characterized by irregular terrain, dense vegetation, and diverse ground materials. Linear probing improves performance but remains insufficient for many terrain categories. In particular, classes such as mud, soft gravel, and rubber surfaces remain poorly segmented. These classes are either absent or underrepresented in common urban datasets, highlighting the need for dedicated benchmarks and improved modeling strategies for unstructured outdoor environments. These observations motivate both the introduction of the EastPark dataset and the development of architectures capable of modeling large-scale contextual dependencies in terrain-dominated scenes.

D. Fully Supervised Training on EastPark

We next evaluate several strong baselines trained from scratch on EastPark, reporting the results in Table II.

LARS-3D achieves the highest segmentation performance among all compared methods, reaching 76.0 mIoU, while requiring only 0.6 GB of GPU memory. This represents a 1.2 mIoU improvement over the baseline, LSK3DNet, with a comparable memory footprint and only a modest increase in latency (82 ms vs. 75 ms). Compared with PTv3, our method improves mIoU by 4.1 points while reducing memory consumption by 85 % (0.6 vs. 4 GB) and latency by 39 % (82 vs. 135 ms). FRNet provides a particularly competitive efficiency baseline, with 0.7 GB memory and 55 ms latency, but drops mIoU by 3.8 points.

Since seq. 05 shares its drive with seqs. 03-04, we also report seq. 02 separately. LARS-3D remains the best method on seq. 02. All methods trained from scratch score substantially higher on seq. 05 (72-88 % mIoU), which shares its drive with training-validation seqs. 03-04.

LARS-3D obtains competitive performance on several key categories, including road, person and car. The main limitation is the terrain category, where it obtains 25.5%, considerably below LSK3DNet and FRNet. Overall, these results demonstrate that LARS-3D provides a favorable accuracy-efficiency trade-off for large-scale outdoor point-cloud segmentation.

E. Results on RELLIS-3D

To evaluate the generalization capability of the proposed architecture, we further conduct experiments on RELLIS-3D, a benchmark designed for off-road autonomous navigation.

Table III reports segmentation performance compared with several existing methods. Our model achieves 45.3 % mIoU, competitive with the best-performing method, Cylinder3D (45.6 %). LARS-3D achieves the highest scores on rubble and barrier and the second-highest on mud, concrete, pole

TABLE II: Semantic segmentation results on the EastPark test set. mIoU is reported on the full test set, on seq. 02 and on seq. 05. Seq. 02 contains no road, sidewalk, rubber, car, bicycle or motorcycle points; its mIoU is averaged over the 14 classes present.

train mode	method	pre-train	**mIoU**	grass	terrain	road	sidew.	person	car	seq 02	seq 05
vanilla	PTv3	nuScenes	5.2	5.2	6.1	6.2	4.3	36.9	17.2	4.2	7.0
	3PNet	SemKITTI	2.7	0.1	5.9	1.6	0.2	21.0	0.7	4.8	0.8
	3PNet	PandaSet	1.5	0.2	1.2	0.9	0.1	0.0	5.3	2.8	1.2
	LSK3DNet	SemKITTI	7.8	14.0	2.3	5.6	1.3	67.5	2.8	11.0	7.4
	LSK3DNet	nuScenes	2.7	20.0	8.2	3.4	1.1	1.3	0.9	2.8	3.7
lin. probing	PTv3	nuScenes	27.1	52.9	1.0	15.8	26.1	75.1	77.9	26.7	24.7
	3PNet	SemKITTI	36.9	73.7	0.1	36.9	38.7	77.6	78.6	35.8	31.9
	3PNet	PandaSet	39.1	72.3	0.3	68.4	43.0	68.8	73.5	35.4	35.2
	LSK3DNet	SemKITTI	26.5	69.6	0.0	5.2	22.5	77.9	64.9	29.5	24.1
	LSK3DNet	nuScenes	28.0	71.1	0.0	20.5	21.8	89.0	78.3	30.4	25.3
train mode	**method**	**VRAM / runtime**									
from scratch	MinkUNet	2 GB / **14 ms**	68.4	84.2	36.7	93.1	90.4	97.9	96.2	58.3	72.6
	3PNet	10 GB / 120 ms	64.2	82.0	19.8	93.6	80.8	94.1	92.7	47.9	72.4
	PTv3	4 GB / 135 ms	71.9	83.2	29.1	94.0	87.2	96.9	95.6	53.8	78.0
	FRNet	0.7 GB / 55 ms	72.2	**85.0**	**42.3**	**97.0**	**93.6**	96.1	95.2	54.5	78.5
	LSK3DNet	**0.6 GB** / 75 ms	74.8	82.7	40.3	95.5	92.8	**98.3**	95.1	59.2	86.5
	LARS-3D	**0.6 GB** / 82 ms	**76.0**	84.0	25.5	96.6	92.5	98.1	**97.1**	**59.8**	**88.0**

TABLE III: Semantic segmentation results on RELLIS-3D test set.

Method	**mIoU**	grass	mud	rubble	pole	barrier
SalsaNext	42.5	65.3	11.5	6.6	44.2	63.6
SPVNAS	42.6	64.1	8.5	5.2	54.4	64.8
GFNet	43.6	64.3	10.1	7.2	49.8	79.3
Xu et al. [47]	44.6	65.8	**12.9**	9.3	50.8	78.4
Cylinder3D	**45.6**	64.9	10.6	2.2	63.7	80.5
MinkUNet	40.7	61.2	5.6	0.5	43.3	62.8
LSK3DNet	43.0	**66.0**	11.6	3.4	**64.8**	74.0
PTv3	32.7	61.2	8.4	1.1	31.8	30.4
LARS-3D	45.3	65.2	11.8	**15.7**	64.4	**84.8**

and fence, while remaining below the best methods on vegetation classes (grass, tree) and log. This confirms that the proposed architecture effectively captures global contextual dependencies, leading to more accurate segmentation in unstructured environments.

F. Results on SemanticPOSS

We further evaluate our method on SemanticPOSS to verify that the proposed architecture maintains strong performance on urban driving datasets. As shown in Table IV, our model achieves 57.3% mIoU, improving over the strongest competing method, LSK3DNet, by 1.5 points. Performance gains are observed across multiple classes, including truck, pole, and trashcan, with notable improvements on small objects and vertical structures. This indicates that the proposed encoder effectively captures contextual relationships between objects and their surroundings, even in densely structured urban scenes. Together with the results on EastPark and RELLIS-3D, these findings demonstrate that the proposed architecture improves segmentation performance across diverse outdoor environments. By integrating GCA modeling, the network can effectively capture global dependencies while maintaining linear computational complexity.

VI. ABLATION STUDY

To evaluate the impact of the proposed GCA encoder, we conduct ablation experiments on the 3 validation sets of the 3 datasets, repeating each experiment 3 times and reporting the average mIoU together with the standard deviation. Table V reports the segmentation performance obtained by progressively replacing the original point branch of LSK3DNet with GCA blocks, and applying different point sorting algorithms. Random and Z-order GCA improve over

TABLE IV: Sem. segmentation results on SemanticPOSS test set.

Method	**mIoU**	person	truck	pole	trash can	ground
SPVCNN	48.4	72.5	31.4	41.3	31.8	75.3
RangeF. [48]	48.9	71.2	27.8	34.6	23.4	77.6
CENet [49]	50.3	75.5	25.3	31.5	48.1	80.3
FRNet	52.5	76.9	28.8	32.9	28.5	79.0
Cylinder3D	52.9	75.9	28.7	37.2	36.7	80.1
MinkUNet	53.1	77.8	33.9	42.5	36.4	79.1
LSK3DNet	55.8	**80.8**	30.4	39.0	49.3	76.5
PTv3	55.4	79.1	33.2	34.2	41.6	**81.4**
LARS-3D	**57.3**	80.3	**34.3**	**43.3**	**59.1**	78.7

TABLE V: Ablation study on the EastPark, RELLIS-3D and SemanticPOSS validation sets (mean ± std over 3 runs). Runtime and VRAM are measured on EastPark.

	Cost		**mIoU (%)**			
Configuration	Time	VRAM	EastPark	RELLIS-3D	SPOSS	Avg
LSK3DNet (baseline)	75 ms	550 MB	65.0 ±0.6	39.6 ±0.7	55.8 ±0.3	53.5 ±0.3
+ Gated linear feature mapping	76 ms	550 MB	64.4 ±0.2	39.5 ±0.5	55.7 ±0.2	53.2 ±0.2
Linear GCA – random order						
+ 2 mixing blocks	78 ms	558 MB	66.7 ±0.6	41.2 ±0.3	55.2 ±0.2	54.4 ±0.2
+ 8 mixing blocks	82 ms	587 MB	67.1 ±0.7	41.8 ±0.2	56.3 ±0.3	55.1 ±0.3
Linear GCA – lexicographic order						
+ 2 mixing blocks	78 ms	558 MB	66.4 ±0.8	42.0 ±1.3	57.2 ±0.1	55.2 ±0.5
+ 8 mixing blocks	82 ms	587 MB	65.6 ±0.6	38.5 ±1.1	55.6 ±0.2	53.2 ±0.4
Linear GCA – Z-order						
+ 2 mixing blocks	78 ms	558 MB	**68.1** ±0.3	42.2 ±0.2	56.3 ±0.3	55.5 ±0.2
+ 8 mixing blocks (**ours**)	82 ms	587 MB	67.3 ±0.5	**42.8** ±0.2	**57.3** ±0.5	**55.8** ±0.2

the baseline at every depth, and Z-order further improves from 2 to 8 blocks (55.5 % to 55.8 %), whereas lexicographic ordering falls below the baseline at 8 blocks. This suggests that the spatial ordering used to linearize the point sequence is important for effective global-context propagation. This represents an average improvement of 2.3 points over the original LSK3DNet baseline at the small cost of increased runtime (7 ms) and Peak VRAM usage (37 MB).

These results demonstrate that the GCA-based encoder effectively enhances the representation capability of the network while maintaining stable training behavior and moderate runtime and VRAM usage.

VII. CONCLUSION

In this work, we investigated LiDAR semantic segmentation in complex outdoor environments characterized by irregular terrain, dense vegetation, and diverse surface types. We found that recent semantic segmentation models trained on urban data struggle to generalize to terrain-rich environments, and we hypothesized that richer global context modeling can help in these scenes. To address this challenge, we introduced a GCA modeling framework for point cloud semantic segmentation that integrates global context reasoning within a sparse convolutional backbone. Our approach replaces the conventional point branch of the SPV architecture with a linear GCA encoder that aggregates global contextual statistics across the point cloud and injects them into point-wise features.

Extensive experiments on three challenging benchmarks demonstrate that the proposed architecture consistently obtains competitive segmentation performance with respect to strong baselines. The ablation study shows that the GCA-based encoder consistently improves over the original backbone when combined with a locality-preserving (Z-order) traversal, at a small runtime and memory cost. Future work will explore temporal and multi-modal perception for richer spatio-temporal reasoning in dynamic outdoor environments.

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